

Lecture 7: REML and MLE

More Advanced Topics

March 6, 2025

From Last Lecture...

- Fitting the maximum likelihood estimator the linear mixed-effects model (LME)
- Fitting the restricted maximum likelihood estimator for the variance parameters in the LME model

High-level Ideas: Mathematical

- Data: $(Y_{ij}, x_{ij1}, \dots, x_{ijp}), i = 1, \dots, m, j = 1, \dots, n_i$

- (A Simple LME) Model:

$$Y_{ij} = \beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + \mathbf{b}_i + \epsilon_{ij}$$

where random effects $b_i \sim N(0, \sigma_b^2)$, independent of $\epsilon_{ij} \sim N(0, \sigma^2)$

- MLE: find $(\beta_1, \dots, \beta_p, \sigma_b, \sigma)$ that maximizes the likelihood function

$$L(\beta_1, \dots, \beta_p, \sigma_b, \sigma) = \prod_{i=1}^m f(Y_{i1}, \dots, Y_{i,n_i})$$

- REML:

(1) find (σ_b, σ) that maximizes the restricted likelihood

$$L_{RE}(\sigma_b, \sigma) = \iiint L(\beta_1, \dots, \beta_p, \sigma_b, \sigma) d\beta_1 \dots d\beta_p$$

(2) With the estimated $(\hat{\sigma}_b, \hat{\sigma})$, find $(\beta_1, \dots, \beta_p)$ that maximizes $L(\beta_1, \dots, \beta_p, \hat{\sigma}_b, \hat{\sigma})$.

High-level Ideas: Computing

- Data: $(Y_{ij}, x_{ij1}, \dots, x_{ijp}), i = 1, \dots, m, j = 1, \dots, n_i$
- Model: $Y_{ij} = \beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + b_i + \epsilon_{ij}$ where $b_i \sim N(0, \sigma_b^2)$, independent of $\epsilon_{ij} \sim N(0, \sigma^2)$
- MLE/REML Algorithm:

- 1) Given $\theta = \frac{\sigma}{\sigma_b}$, find $(\beta_1, \dots, \beta_p)$ and $(b_1, \dots, b_m)$ that minimizes

$$RSS(\theta) = \sum_{i=1}^m \left(\sum_{j=1}^{n_i} (Y_{ij} - (\beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + b_i))^2 + \theta^2 b_i^2 \right)$$

- 2) The REML estimate $\hat{\sigma}_\theta^2 = \frac{RSS(\theta)}{N-p}$ and MLE estimate $\hat{\sigma}_\theta^2 = \frac{RSS(\theta)}{N}$
- 3) The REML finds θ that maximizes $L_{RE}(\theta) = L_{RE}(\hat{\sigma}_\theta/\theta, \hat{\sigma}_\theta)$.
The MLE finds θ that maximizes the likelihood $L(\hat{\beta}(\theta), \hat{\sigma}_\theta/\theta, \hat{\sigma}_\theta)$

A comparison of MLE and REML

- MLE and REML estimators will often give very similar results
- When the number p of fixed effects parameters is **relatively large** as compared to the total number N of observations, then they could be very different.
 - In that case, the REML should be less biased
 - In general, REML is better for inference on parameters
- A main advantage of the MLE is that it allows us to apply AIC or BIC or log-likelihood criterion for model selection or comparison!!!
 - In general, MLE is better for model selection

Computing Issues of MLE and REML

Both REML and MLE might not converge and have warning messages:

- The optimal solution is not unique when minimizing

$$RSS(\theta) = \sum_{i=1}^m \left(\sum_{j=1}^{n_i} (Y_{ij} - (\beta_1 x_{ij1} + \cdots + \beta_p x_{ijp} + b_i))^2 + \theta^2 b_i^2 \right)$$

e.g., the standard linear regression $Y \sim X_1 + \cdots + X_p$ has infinite solutions (due to collinearity).

- When minimizing likelihood function $L\left(\hat{\beta}(\theta), \frac{\hat{\sigma}_\theta}{\theta}, \hat{\sigma}_\theta\right)$, or restricted likelihood $L_{RE}(\theta)$, the optimization algorithm of estimating θ might diverge, e.g., $\theta \rightarrow 0$ or ∞ .

(1) Ball Weight Problems from Week#6

- Two NYU students, “Jack” and “Kathy”, work together to find the weights of two balls, A and B, using a scale with random measurement errors. Each student took two measurements, and
- “Jack” measures ball “A” and “B” each one time, and the observed weights are 2 and 4 lbs.
- “Kathy” measures “B” and “A+B”, and observed weights are 3 and 5 lbs.

Questions: use the linear mixed effect model to find the weights of balls A and B.

Data → Model

	A	B	A+B
Jack	2	4	--
Kathy	--	3	5

- **Model: Observed Data = True Value + Noise** where
Noise = Subject-Random-Effects + Measurement Error

$$2 = Y_{11} = A + b_1 + \epsilon_{11}$$

$$4 = Y_{12} = B + b_1 + \epsilon_{12}$$

$$3 = Y_{21} = B + b_2 + \epsilon_{21}$$

$$5 = Y_{22} = A + B + b_2 + \epsilon_{22}$$

R code

```
data1 <- rbind( c(1,2,1,0), c(1,4,0,1), c(2,3, 0,1), c(2,5,1,1) );  
colnames(data1) <- c("ID", "Y", "X1", "X2");  
data1 = data.frame(data1);
```

REML

```
lme1 <- lmer( Y ~ X1 + X2 - 1 + (1 | ID), data= data1);  
summary(lme1)
```

MLE

```
lme2 <- lmer( Y ~ X1 + X2 - 1 + (1 | ID), REML=F, data= data1);  
summary(lme2)
```

```
> summary(lme1)
Linear mixed model fit by REML ['lmerMod']
Formula: Y ~ X1 + X2 - 1 + (1 | ID)
Data: data1
```

R Output for REML

REML criterion at convergence: 4.1

Scaled residuals:

Min	1Q	Median	3Q	Max
-0.6708	-0.3354	0.0000	0.2795	0.4472

$$\hat{\theta} = \frac{0.2236}{0.5000} = 0.4472$$

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	0.25	0.5000
	Residual	0.05	0.2236

Number of obs: 4, groups: ID, 2

Fixed-effect parameter from REML

```
> fixef(lme1)
X1 X2
1.80 3.65
```

Fixed effects:

	Estimate	Std. Error	t value
X1	1.8000	0.2345	7.675
X2	3.6500	0.2872	12.708

```
> ranef(lme1)
$ID
(Intercept)
```

1	0.25
2	-0.50

Correlation of Fixed Effects:

X1	
X2	0.408

R code for linear regression

```
theta = sqrt(0.05/0.25) ##0.4472
```

```
## new data
```

```
data2 <- rbind( c(2,1,0, 1,0), c(4,0,1, 1,0), c(0,0,0,theta,0),  
               c(3,0,1, 0, 1), c(5,1,1, 0, 1), c(0,0,0,0, theta) );
```

```
colnames(data2) <- c( "Y", "A", "B", "b1", "b2" );
```

```
data2 <- data.frame(data2)
```

```
# standard regression on new data
```

```
lm(Y ~ . -1, data=data2)
```

Call:

```
lm(formula = Y ~ . - 1, data = data2)
```

Coefficients:

A	B	b1	b2
1.80	3.65	0.25	-0.50

Optional: REML function

- **Fact:** for $RSS(A, B) = \sum_{i=1}^K (Y^* - u_i A - v_i B)^2$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{RSS(A,B)}{2\sigma^2}} dA dB = \left(\sqrt{2\pi}\sigma\right)^2 \frac{1}{\sqrt{\Delta}} e^{-\frac{\min_{A,B} RSS}{2\sigma^2}}$$

where

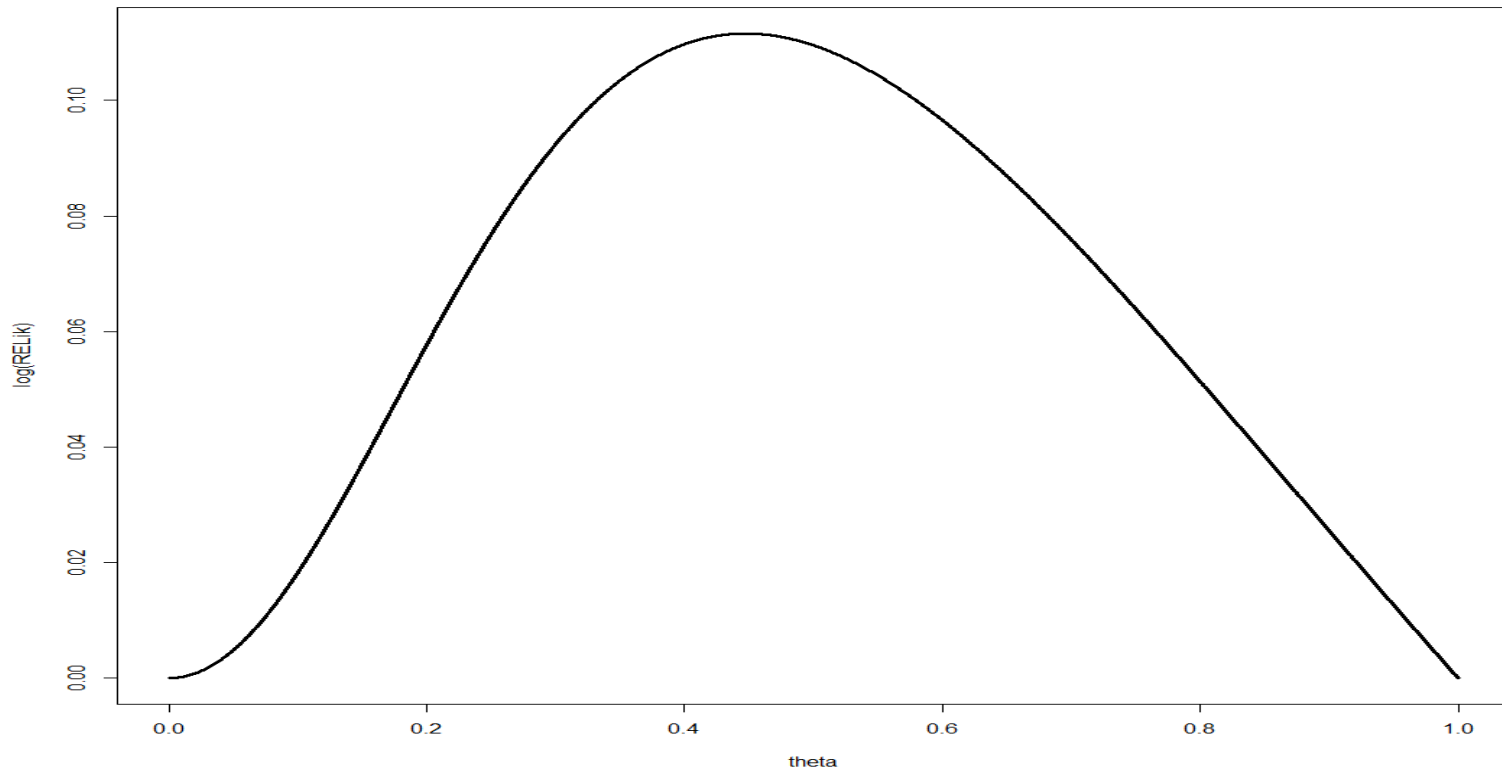
$$\Delta = (u_1^2 + \dots + u_K^2)(v_1^2 + \dots + v_K^2) - (u_1 v_1 + \dots + u_K v_K)^2$$

- For this particular dataset,

$$\begin{aligned} L_{RE}(\theta) = L_{RE}(\sigma_b, \sigma) &= \frac{\theta^2}{\left(\sqrt{2\pi} \sigma\right)^2} \frac{1}{\sqrt{5\theta^4 + 10\theta^2 + 1}} e^{-\frac{RSS(\theta)}{2\sigma^2}} \\ &= \frac{\theta^2}{\pi * RSS(\theta)} \frac{1}{\sqrt{5\theta^4 + 10\theta^2 + 1}} e^{-1} \end{aligned}$$

Optional: REML

Below is the plot of $\text{Log}(RE\text{Lik}(\theta))$ as a function of θ !
This yield the REML estimate of $\hat{\theta}_{REML} = 0.4472$.



Optional: R code for plot

```
theta <- seq(0.0001,1,0.0001);
# new data
RELik <- NULL;
for (i in 1:length(theta)){
  data2 <- rbind( c(2,1,0, 1,0),  c(4,0,1, 1,0), c(0,0,0,theta[i],0),
                c(3,0,1, 0, 1),  c(5,1,1, 0, 1), c(0,0,0,0, theta[i]) );
  colnames(data2) <- c( "Y", "A", "B", "b1", "b2" );
  data2 <- data.frame(data2)
  # standard regression on new data
  lm2 <- lm(Y ~ . -1, data=data2);
  # RSS values
  RSS2 <- deviance(lm2);
  ##restricted likelihood function
  liktmp <- (theta[i]^2) *exp(-1) / (pi* sqrt(5*theta[i]^4+10*theta[i]^2+1) * RSS2);
  RELik <- c(RELik, liktmp ) ;
}
plot(theta, log(RELik),"l", lwd=3)
theta[which.max(RELik)]    #0.4472
```

Optional: REML of variance parameters

```
theta.REML = theta[which.max(RELik)];  
data2.REML <- rbind( c(2,1,0, 1,0),  c(4,0,1, 1,0), c(0,0,0,theta.REML,0),  
                    c(3,0,1, 0, 1),  c(5,1,1, 0, 1), c(0,0,0,0, theta.REML) );  
colnames(data2.REML) <- c( "Y", "A", "B", "b1", "b2" );  
data2.REML <- data.frame(data2.REML)  
lm2.REML <- lm(Y ~ . -1, data=data2.REML);
```

Find the estimate of sigma

```
sigma.REML <- sqrt(deviance(lm2.REML)/ 2);
```

find the estimate of sigma.b

```
sigma.b.REML <- sigma.REML/theta.REML;
```

Results

```
round(c(sigma.REML, sigma.b.REML), 4)  ##[1] 0.2236 0.5000
```

MLE in LME

MLE: find $(\beta_1, \dots, \beta_p, \sigma_b, \sigma)$ that maximizes the likelihood function

$$L(\beta_1, \dots, \beta_p, \sigma_b, \sigma) = \prod_{i=1}^m f(Y_{i1}, \dots, Y_{i,n_i})$$

MLE in R

```
lme2 <- lmer( Y ~ X1 + X2 - 1 + (1 | ID), REML=F, data= data1);  
summary(lme2)
```


R Output for MLE

```
> lme2 <- lmer( Y ~ X1 + X2 - 1 + (1 | ID), REML=F, data= data1);
```

Warning messages:

- 1: In optwrap(optimizer, devfun, getStart(start, rho\$pp), lower = rho\$lower, :
convergence code -4 from nloptwrap: NLOPT_ROUNDOff_LIMITED:
Roundoff errors led to a breakdown of the optimization algorithm. In this case, the returned minimum may still be useful. (e.g. this error occurs in NEWUOA if one tries to achieve a tolerance too close to machine precision.)
 - 2: In checkConv(attr(opt, "derivs"), opt\$par, ctrl = control\$checkConv, :
unable to evaluate scaled gradient
 - 3: In checkConv(attr(opt, "derivs"), opt\$par, ctrl = control\$checkConv, :
Model failed to converge: degenerate Hessian with 1 negative eigenvalues
- >
- >

R output for MLE

```
> summary(lme2)
Linear mixed model fit by maximum likelihood ['lmerMod']
Formula: Y ~ X1 + X2 - 1 + (1 | ID)
Data: data1
```

```
    AIC    BIC logLik deviance df.resid
-59.3  -61.7  33.6  -67.3      0
```

Scaled residuals:

Min	1Q	Median	3Q	Max
-6.517e-07	-1.629e-07	1.901e-07	4.481e-07	6.517e-07

Random effects:

Groups	Name	Variance	Std.Dev.
ID	(Intercept)	2.500e-01	5.000e-01
Residual		1.672e-17	4.088e-09

Number of obs: 4, groups: ID, 2

Fixed effects:

	Estimate	Std. Error	t value
X1	2.000e+00	5.782e-09	345902118
X2	4.000e+00	8.177e-09	489179467

Correlation of Fixed Effects:

X1	
X2	0.707

optimizer (nloptwrap) convergence code: -4 (NLOPT_ROUNDOff_LIMITED: Roundoff errors led to a breakdown of the optimization algorithm. In this case, the returned minimum may still be useful. (e.g. this error occurs in NEWUOA if one tries to achieve a tolerance too close to machine precision.))

unable to evaluate scaled gradient

Model failed to converge: degenerate Hessian with 1 negative eigenvalues

Parameter from MLE

```
> fixef(lme2)
```

X1	X2
2	4

```
> ranef(lme2)
```

```
$ID
```

```
  (Intercept)
```

```
1 -6.908407e-15
```

```
2 -1.000000e+00
```

MLE: R code for linear regression

```
theta = sqrt(1.672e-17/0.25)
## new data
data3 <- rbind( c(2,1,0, 1,0),  c(4,0,1, 1,0), c(0,0,0,theta,0),
c(3,0,1, 0, 1),  c(5,1,1, 0, 1), c(0,0,0,0, theta) );

colnames(data3) <- c( "Y", "A", "B", "b1", "b2" );
data3 <- data.frame(data3)

# standard regression on new data
lm(Y ~ . -1, data=data3)
```

MLE: R code from Linear Regression

```
> lm(Y ~ . -1, data=data3)
```

Call:

```
lm(formula = Y ~ . - 1, data = data3)
```

Coefficients:

A	B	b1	b2
---	---	----	----

2.000e+00	4.000e+00	-3.753e-16	-1.000e+00
-----------	-----------	------------	------------

Potential Issue when $\theta \rightarrow 0$

- What happens if $\hat{\theta} \rightarrow 0$?

$$RSS(\theta) = \sum_{i=1}^m \left(\sum_{j=1}^{n_i} (Y_{ij} - (\beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + b_i))^2 + \theta^2 b_i^2 \right)$$

- Derived data for linear regression:

	Y	X1	X2	b1	b2
1	2	1	0	1	0
2	4	0	1	1	0
3	0	0	0	0	0
4	3	0	1	0	1
5	5	1	1	0	1
6	0	0	0	0	0

- Perfect fit:** $\hat{A} = 2, \hat{B} = 4, \hat{b}_1 = 0, \hat{b}_2 = -1, RSS = 0!!!$
Thus the MLE of $\hat{\theta} = 0!!!$

Optional: MLE in LME

- Fact:** for $RSS(A, B, b) = \sum_{i=1}^K (Y^* - u_i A - v_i B - w_i b)^2$

$$\int_{-\infty}^{\infty} e^{-\frac{RSS(A, B, b)}{2\sigma^2}} db = \sqrt{2\pi}\sigma \frac{1}{\sqrt{w_1^2 + \dots + w_K^2}} e^{-\frac{\min_b RSS}{2\sigma^2}}$$

- MLE:** for this particular dataset, find $(\beta_1, \dots, \beta_p, \sigma_b, \sigma)$ that maximizes the **likelihood function**

$$L(\theta) = L(\beta_1, \dots, \beta_p, \sigma_b, \sigma) = \prod_{i=1}^m f(Y_{i1}, \dots, Y_{i,n_i})$$

$$= \frac{\theta^2}{(\sqrt{2\pi})^4 \sigma^4 (\theta^2 + 2)} \exp\left(-\frac{RSS(\theta)}{2\sigma^2}\right) = \frac{16 \theta^2}{(\sqrt{2\pi})^4 (\theta^2 + 2) [RSS(\theta)]^2} \exp(-2)$$

- A comparison with the REML function:**

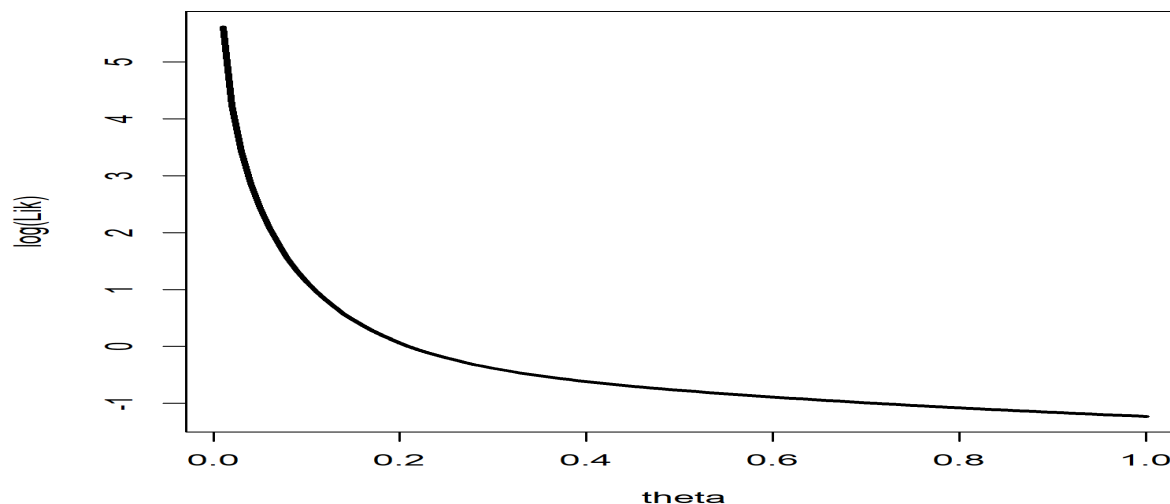
$$L_{RE}(\theta) = \frac{\theta^2}{(\sqrt{2\pi})^2 \sigma^2} \frac{1}{\sqrt{5\theta^4 + 10\theta^2 + 1}} e^{-\frac{RSS(\theta)}{2\sigma^2}} = \frac{\theta^2}{\pi \sqrt{5\theta^4 + 10\theta^2 + 1} [RSS(\theta)]} \exp(-1)$$

What happens to the likelihood function?

- For this dataset, the MLE $\hat{\sigma}^2 = RSS(\theta)/4$, and thus

$$\begin{aligned} \text{Lik}(\theta) &= L(\beta_1, \dots, \beta_p, \sigma_b, \sigma) = \prod_{i=1}^m f(Y_{i1}, \dots, Y_{i,n_i}) \\ &= \frac{\theta^2}{(\sqrt{2\pi})^4 \sigma^4 (\theta^2 + 2)} \exp\left(-\frac{RSS(\theta)}{2\sigma^2}\right) = \frac{16 \theta^2}{(\sqrt{2\pi})^4 (\theta^2 + 2) [RSS(\theta)]^2} \exp(-2) \end{aligned}$$

- Below is the plot of $\text{Loglik}(\theta)$ as a function of θ !



R code to generate the plot

```
theta <- seq(0.01,1,0.01);

# new data
Lik <- NULL;
for (i in 1:length(theta)){
  data2 <- rbind( c(2,1,0, 1,0), c(4,0,1, 1,0), c(0,0,0,theta[i],0),
                 c(3,0,1, 0, 1), c(5,1,1, 0, 1), c(0,0,0,0, theta[i]) );
  colnames(data2) <- c( "Y", "A", "B", "b1", "b2" );
  data2 <- data.frame(data2)

# standard regression on new data
lm2 <- lm(Y ~ . -1, data=data2);

# RSS values. Another way to use # sum(resid(lm2)^2)
RSS2 <- deviance(lm2);
## MLE of sigma2
sigma2 <- sqrt(RSS2 / 4);

##likelihood function
if (sigma2 == 0) liktmp = 0
else {
  liktmp <- (theta[i]^2 / (theta[i]^2 + 2)) * (1/ (2*pi)^2) * (1/(sigma2)^4) * exp(-RSS2/ (2*sigma2^2));
}
Lik <- c(Lik, liktmp ) ;
}
plot(theta, log(Lik),"l", lwd=3)
```


Example #2: Ball weight in Week #4

- Suppose three other students, “David”, “Emily” and “Frank”, were also asked to work together to find the weights of two balls, A and B, using a scale with random measurement errors. However, their team strategies are different.
- “David ” measured the weight of ball “A” three times, and the observed weights are 1,2,3 lbs.
- “Emily” measured the weight of ball “B” three times, and the observed weights are 4,2,3 lbs.
- “Frank” measured the combined weight of “(A+B)” three times, and the observed weights are 4,3,5 lbs.

Questions: find the best estimates of the weights of balls A and B.

Data → Model

David	Emily	Frank
A	B	A+B
1	4	4
2	2	3
3	3	5

- Model for David/Emily/Frank

$$1 = A + b_1 + \epsilon_{11}$$

$$2 = A + b_1 + \epsilon_{12}$$

$$3 = A + b_1 + \epsilon_{13}$$

$$4 = B + b_2 + \epsilon_{21}$$

$$2 = B + b_2 + \epsilon_{22}$$

$$3 = B + b_2 + \epsilon_{23}$$

$$4 = A + B + b_3 + \epsilon_{31}$$

$$3 = A + B + b_3 + \epsilon_{32}$$

$$5 = A + B + b_3 + \epsilon_{33}$$

Assumptions:

- b_1, b_2, b_3 are iid $N(0, \sigma_b^2)$
- ϵ_{ij} are iid $N(0, \sigma^2)$

Key parameter

$$\theta = \frac{\sigma_b}{\sigma}$$

R code & output from Week #4

```
> summary(lme02 <- lmer(Y ~ X1 + X2-1 + (1 | ID2), data = data0))  
Linear mixed model fit by REML ['lmerMod']  
Formula: Y ~ X1 + X2 - 1 + (1 | ID2)  
Data: data0
```

REML criterion at convergence: 23.2

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.3333	-0.6667	0.3333	0.6667	1.3333

Random effects:

Groups	Name	Variance	Std.Dev.
ID2	(Intercept)	1.745e-08	0.0001321
	Residual	1.000e+00	1.0000000

Number of obs: 9, groups: ID2, 3

Fixed effects:

	Estimate	Std. Error	t value
X1	1.6667	0.4714	3.536
X2	2.6667	0.4714	5.657

Correlation of Fixed Effects:

X1
X2 -0.500

optimizer (nloptwrap) convergence code: 0 (OK)

Hessian is numerically singular: parameters are not uniquely determined

Warning message:

In checkConv(attr(“opt”, “derivs”), opt\$par, ctrl = control\$checkConv, :
Hessian is numerically singular: parameters are not uniquely determined

For David/Emily/Frank:

REML:

$$\hat{\theta} = 7570.023$$

R code & output for **MLE**

```
> summary(lme03 <- lmer(Y ~ X1 + X2 - 1 + (1 | ID2), REML=F, data = data0))
```

boundary (singular) fit: see help('isSingular')

Linear mixed model fit by maximum likelihood [lmerMod']

Formula: Y ~ X1 + X2 - 1 + (1 | ID2)

Data: data0

AIC	BIC	logLik	deviance	df.resid
-----	-----	--------	----------	----------

31.3	32.1	-11.6	23.3	5
------	------	-------	------	---

Scaled residuals:

Min	1Q	Median	3Q	Max
-1.5119	-0.7559	0.3780	0.7559	1.5119

Random effects:

Groups	Name	Variance	Std.Dev.
--------	------	----------	----------

ID2	(Intercept)	0.0000	0.0000
-----	-------------	--------	--------

Residual		0.7778	0.8819
----------	--	--------	--------

Number of obs: 9, groups: ID2, 3

Fixed effects:

	Estimate	Std. Error	t value
X1	1.6667	0.4157	4.009
X2	2.6667	0.4157	6.414

Correlation of Fixed Effects:

X1	
X2	-0.500

optimizer (nloptwrap) convergence code: 0 (OK)

boundary (singular) fit: see help('isSingular')

For David/Emily/Frank:

MLE:

$$\hat{\theta} = \infty$$

Potential Issue when $\hat{\theta} \rightarrow \infty$

- What happens if $\hat{\theta} \rightarrow \infty$? **or** $\hat{\sigma}_b = \frac{\hat{\sigma}}{\hat{\theta}} = \mathbf{0!!!}$

$$RSS(\theta) = \sum_{i=1}^m \left(\sum_{j=1}^{n_i} (Y_{ij} - (\beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + b_i)) \right)^2 + \theta^2 b_i^2$$

$\rightarrow b_1 = \dots = b_m = \mathbf{0!}$ No need to run linear mixed-effects models!!!

- A comparison with the following R outputs (**REML** and **MLE**):

```
> fixef(lme02)
```

X1	X2
----	----

1.666667	2.666667
----------	----------

```
> fixef(lme03)
```

X1	X2
----	----

1.666667	2.666667
----------	----------

```
> ranef(lme02)
```

\$ID2

(Intercept)

David 1.745339e-08

Emily 1.745339e-08

Frank -1.745339e-08

with conditional variances for "ID2"

```
> ranef(lme03)
```

\$ID2

(Intercept)

David 0

Emily 0

Frank 0

with conditional variances for "ID2"

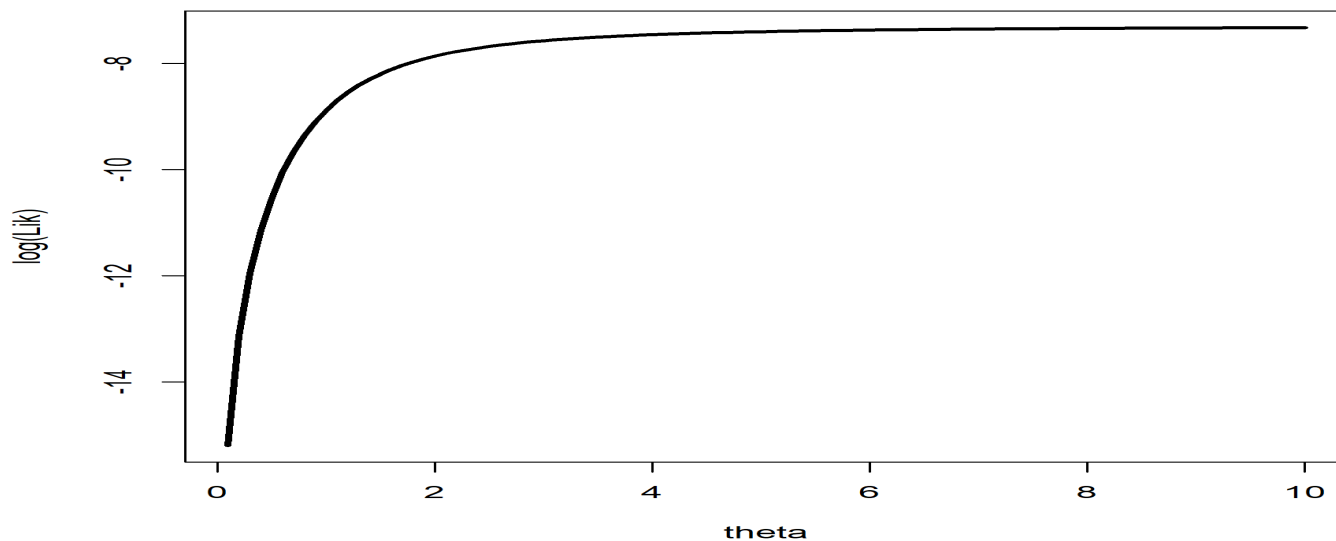
What happens to the likelihood function?

- For this dataset, the MLE $\hat{\sigma}^2 = \frac{RSS(\theta)}{N=9}$ and thus

$$Lik(\theta) = L(\beta_1, \dots, \beta_p, \sigma_b, \sigma) = \prod_{i=1}^m f(Y_{i1}, \dots, Y_{i,n_i})$$

$$= \frac{\theta^3}{(\sqrt{2\pi})^9 (\sqrt{\theta^2 + 3})^3 \sigma^9} \exp\left(-\frac{RSS(\theta)}{2\sigma^2}\right) = \frac{\theta^3 9^{9/2}}{(\sqrt{2\pi})^9 (\sqrt{\theta^2 + 3})^3 [RSS(\theta)]^{9/2}} \exp\left(-\frac{9}{2}\right)$$

- Below is the plot of $Loglik(\theta)$ as a function of θ !



R code for the plot

```
theta <- seq(0.1,10,0.1);

# new data
Lik <- NULL;
for (i in 1:length(theta)){
  data6 <- rbind( c(1,1,0, 1,0, 0), c(2,1,0, 1,0,0), c(3,1,0,1,0,0), c(0,0,0,theta[i],0,0),
    c(4,0,1, 0,1, 0), c(2,0,1, 0,1,0), c(3,0,1,0,1,0), c(0,0,0,0, theta[i],0),
    c(4,1,1, 0,0, 1), c(3,1,1, 0,0,1), c(5,1,1,0,0,1), c(0,0,0,0,0,theta[i]) );
  colnames(data6) <- c( "Y", "A", "B", "b1", "b2", "b3" );
  data6 <- data.frame(data6)

  # standard regression on new data
  lm6 <- lm(Y ~ . -1, data=data6);

  # RSS values. Another way to use
  # sum(resid(lm6)^2)
  RSS6 <- deviance(lm6);
  ## MLE of sigma2
  sigma6 <- sqrt(RSS6 / 9);

  ##likelihood function
  if (sigma6 == 0) liktmp = 0
  else {
    liktmp <- (theta[i]^3 / (theta[i]^2 + 3)^(3/2)) * (1/ (2*pi)^(9/2)) * (1/(sigma6)^9) * exp(-RSS2/ (2*sigma6^2));
  }
  Lik <- c(Lik, liktmp) ;
}
plot(theta, log(Lik),"l", lwd=3)
```

What if there is a warning message?

When running REML or MLE to fit linear mixed-effects model, how to proceed if there is a warning message

- It is likely that $\hat{\theta} \rightarrow 0$ or ∞ .
- The variance estimates of σ and σ_b can be problematic (e.g., one of them is 0). More generally, in the model with q random effects, the estimate of the covariance structure of $b_i = (b_{i1}, \dots, b_{iq})$ would be problematic.
- You can get more insights by checking
$$RSS(\theta) = \sum_{i=1}^m \left(\sum_{j=1}^{n_i} (Y_{ij} - (\beta_1 x_{ij1} + \dots + \beta_p x_{ijp} + b_i)) \right)^2 + \theta^2 b_i^2$$
and the (profile) likelihood function of θ [or $h(\theta)/\sigma^{N-p}$ for REML] is
$$Lik(\theta) = L(\beta_1, \dots, \beta_p, \sigma_b, \sigma) = \prod_{i=1}^m f(Y_{i1}, \dots, Y_{i,n_i}) = \frac{g(\theta)}{(\sqrt{2\pi})^N \sigma^N} \exp\left(-\frac{RSS(\theta)}{2\sigma^2}\right)$$
- The point estimator of $(\beta_1, \dots, \beta_p)$ might (or might not) still be meaningful, but the hypothesis testing or confidence intervals will likely be problematic.

Summary

- **REML and MLE will often give very similar results**, esp. if the number p of parameters is small.
- **When there are warning messages**, the algorithm of estimating the variance component likely does not converge to a meaningful value, e.g., $\theta = \sigma/\sigma_b \rightarrow 0$ or ∞ .
- **The REML is generally better for parameter inference**, as it often yields unbiased estimators, esp. when the number p of parameters is large as compared to the number of observations.
- **The MLE is generally better for model selection or comparison** via AIC/BIC/log-likelihood criterion
 - REML cannot be used for AIC/BIC criterion.